MSc Data Science

DS601: Mathematical Foundations for Data Science

Tutorial 3 - Complete Worked Solutions

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Question 1: Online Examination with Two Attempts

60% of students clear Attempt 1 and 40% fail it. Of those who clear Attempt 1, 90% are finally selected. Of those who fail Attempt 1, 70% appear for Attempt 2, and 80% of those who appear for Attempt 2 are finally selected. Given that a randomly selected student was finally selected, find the probability that the student had failed Attempt 1.

1. Concept Used

- Conditional probability
- Law of Total Probability
- Bayes' Theorem

Bayes' theorem is used because we know the final outcome (the student was selected) and want to work backward to determine which earlier path the student followed.

$$P(A \mid B) = P(A \cap B) \,/\, P(B)$$

2. Approach

There are two mutually exclusive paths to final selection:

- Clear Attempt 1 -> selected.
- Fail Attempt 1 -> appear for Attempt 2 -> selected.

Find the probability of each path, add them to obtain the total probability of selection, and then divide the failed-Attempt-1 selection path by the total selected probability.

3. Step-by-Step Solution

Let C = clear Attempt 1, F = fail Attempt 1, and S = finally selected.

$$P(C) = 0.60, P(F) = 0.40$$

Selection through Attempt 1:

$$P(C \cap S) = P(C)\, P(S \mid C) = 0.60 \times 0.90 = 0.54$$

For a student who failed Attempt 1:

$$P(S \mid F) = P(\text{appear for Attempt 2} \mid F) \times P(S \mid \text{appear for Attempt 2})$$

$$P(S \mid F) = 0.70 \times 0.80 = 0.56$$

$$P(F \cap S) = P(F)\, P(S \mid F) = 0.40 \times 0.56 = 0.224$$

Total probability of final selection:

$$P(S) = 0.54 + 0.224 = 0.764$$

Now apply Bayes' theorem:

$$P(F \mid S) = P(F \cap S) / P(S) = 0.224 / 0.764 = 56/191 \approx 0.2932$$

Question 2: Ten-Door Monty Hall Problem

There are 10 closed doors: one contains a car and nine contain goats. You choose Door 1. The host knows the car's location and opens eight other doors, all containing goats. Only Door 1 and Door 7 remain closed. Find the probability of winning by switching, decide whether to stay or switch, and explain whether 200 previous games change the next game's probability.

1. Concept Used

- Conditional probability and information revealed by the host
- Generalized Monty Hall reasoning
- Independence of repeated games
- Gambler's fallacy

2. Approach

At the instant of the first choice, Door 1 has probability 1/10 of containing the car. The remaining nine doors collectively have probability 9/10. Because the host deliberately opens only goat doors, revealing eight goats does not transfer probability to Door 1. Instead, the entire 9/10 probability of the other nine doors is concentrated on the only other unopened door.

3. Step-by-Step Solution

$$P(\text{car behind Door 1}) = 1/10$$

$$P(\text{car somewhere among the other 9 doors}) = 9/10$$

After the host opens eight goat doors, Door 7 is the only unopened door from that group of nine.

$$P(\text{car behind Door 7}) = 9/10$$

(i) Probability of winning by switching

$$P(\text{win by switching}) = 9/10 = 0.90$$

Answer: 90%.

(ii) Which strategy?

$$\text{Stay: } 1/10 = 10\% \text{ Switch: } 9/10 = 90\%$$

Answer: Switch.

(iii) Does playing 200 previous games change the next game's probability? No. If each game is set up independently, the location of the car in the next game does not depend on what happened in previous games. Therefore the probability that the initially chosen Door 1 contains the car remains 1/10 in every new game.

Believing that a particular outcome becomes more likely merely because it has not occurred recently is the gambler's fallacy.

$$P(\text{Door 1 wins on next independent game}) = 1/10$$

Final Answer: Previous games do not increase the 1/10 probability.

Question 3: Rare Disease and Two Positive Tests

A disease affects 1% of the population. A diagnostic test has 98% sensitivity and a 3% false-positive rate among healthy people. A person receives two positive results from two independent tests having the same characteristics. Find the probability that the person actually has the disease.

1. Concept Used

- Bayes' Theorem
- Conditional probability
- Independence of the two tests conditional on disease status

2. Approach

Consider two possible states: diseased (D) and healthy (D^c). Compute the probability of two positive tests in each state. Because the two tests are independent once the person's disease status is fixed, multiply the two positive-test probabilities. Then use Bayes' theorem.

3. Step-by-Step Solution

$$P(D) = 0.01, P(D^c) = 0.99$$

$$P(+ \mid D) = 0.98, P(+ \mid D^c) = 0.03$$

Two positive results if the person is diseased:

$$P(++ \mid D) = 0.98 \times 0.98 = 0.9604$$

$$P(D \cap ++) = P(D) \times P(++ \mid D) = 0.01 \times 0.9604 = 0.009604$$

Two positive results if the person is healthy:

$$P(++ \mid D^c) = 0.03 \times 0.03 = 0.0009$$

$$P(D^c \cap ++) = 0.99 \times 0.0009 = 0.000891$$

Total probability of two positive tests:

$$P(++) = 0.009604 + 0.000891 = 0.010495$$

Apply Bayes' theorem:

$$P(D \mid ++) = 0.009604 / 0.010495 = 9604/10495 \approx 0.9151$$

Final Answer: P(disease | two positive tests) ≈ 91.51%.

Question 4: Production Plants and Automated Inspection

Plants P1, P2, and P3 produce 30%, 45%, and 25% of total output. Their defect probabilities are 0.02, 0.03, and 0.05. The inspection detects a defective item with probability 0.95 and falsely identifies a non-defective item as defective with probability 0.04. Answer parts (a)-(c).

1. Concept Used

- Law of Total Probability
- Bayes' Theorem
- True positive and false positive events

● Complement rule and independence for part (c)

2. Approach

First find the overall probability that an item is actually defective. Then distinguish carefully between D = actually defective and I = inspection says defective. For part (c), condition on the fact that each item has already been identified as defective, and use the complement of the event 'both are actually non-defective'.

3. Step-by-Step Solution

Given:

$$P(P1)=0.30,\ P(P2)=0.45,\ P(P3)=0.25$$

$$P(D|P1)=0.02,\ P(D|P2)=0.03,\ P(D|P3)=0.05$$

$$P(I|D)=0.95,\ P(I|D^c)=0.04$$

Part (a): P(P3 | D)

First find the overall defect probability:

$$P(D)=0.30(0.02)+0.45(0.03)+0.25(0.05)$$

$$P(D)=0.006+0.0135+0.0125=0.032$$

The probability of 'P3 AND defective' is:

$$P(P3 \cap D)=0.25(0.05)=0.0125$$

Bayes' theorem gives:

$$P(P3 \mid D)=0.0125/0.032=0.390625$$

Answer (a): approximately 39.06%.

Part (b):

Concept: Bayes' theorem with an imperfect test. Important difference:

Step 1 — Actually defective items.

Total defective = 32.

Inspection catches 95%:

32×0.95 = 30.4.

Step 2 — Healthy items.

1000−32 = 968 healthy.

Step 3 — Healthy items false positives: 4% of 968:

968×0.04 = 38.72.

Inspection comes from two groups

Actually defective + correctly detected = 30.4

Actually healthy + wrongly detected = 38.72

Total inspection-positive = 30.4 + 38.72 = 69.12.

Step 5 — P3 contribution in inspection ?

P3 actual defective = 12.5×0.95 = 11.875.

Step 6 — Ab ratio.

P(P3|I) = 11.875/69.12 = 0.1718.

Answer (b): 17.18%

Part (c): Two flagged items, at least one truly defective

First determine the probability that one flagged item is truly defective:

$$P(D \mid I)=P(D)P(I|D)/P(I)=0.0304/0.06912=95/216\approx 0.439815$$

$$P(D^c \mid I)=1-95/216=121/216\approx 0.560185$$

The two selected items are independent. Therefore, conditional on both being flagged, the probability that both are actually non-defective is:

$$P(\text{both actually non-defective} \mid \text{both flagged})=(121/216)^2$$

Use the complement rule:

$$P(\text{at least one actually defective} \mid \text{both flagged})=1-(121/216)^2$$

$$=32015/46656\approx 0.68619$$

Answer (c): approximately 68.62%.

Question 5: Ten Doors with Three Doors Left Closed

There are 10 doors. You choose Door 1. The host knows the car's location and opens six goat doors. Three doors remain closed: Door 1, Door 7, and Door 10. Find the probability that Door 1 has the car, the collective probability for Doors 7 and 10, the winning probability if you choose randomly between Doors 7 and 10, and whether switching is still better.

1. Concept Used

- Generalized Monty Hall problem
- Conditional probability
- Probability carried by a set of alternatives

2. Approach

Your original door begins with probability 1/10. The other nine doors collectively begin with probability 9/10. The host opens only goat doors, so the probability attached to your original choice remains 1/10, while the 9/10 probability of the other group is now concentrated on the two unopened alternative doors.

3. Step-by-Step Solution

Part (a): Door 1

$$\text{P(car behind Door 1)} = 1/10 = 0.10$$

Answer (a): 10%.

Part (b): Door 7 or Door 10 collectively

$$\text{P(car behind Door 7 OR Door 10)} = 9/10 = 0.90$$

Answer (b): 90%.

Part (c): Randomly choose between Door 7 and Door 10

You choose each of the two remaining alternative doors with probability 1/2. The total probability that the car is somewhere in that pair is 9/10. Therefore the random-switch strategy wins with:

$$\text{P(win)} = 1/2 \times 9/10 = 9/20 = 0.45$$

Answer (c): 45%.

Part (d): Is switching still better?

$$\text{Stay} = 10\% \quad \text{Random switch among the two alternatives} = 45\%$$

Answer (d): Yes. Switching is still better than staying.

Question 6: Simpson's Paradox - Easy Numerical Method

This question compares Treatment A and Treatment B for Mild (M) and Severe (S) patients. We first calculate the overall recovery rates, then find when Simpson's Paradox occurs, and finally show why randomization removes the reversal.

Numerical values used for this Q6 solution: Treatment A recovers 90% of Mild patients and 50% of Severe patients; Treatment B recovers 80% of Mild patients and 30% of Severe patients. For Treatment A, alpha is the Mild-patient proportion. For the numerical Treatment B aggregate, the patient mix used is 80% Mild and 20% Severe.

1. Concept Used

- Law of Total Probability: overall recovery is a weighted average of Mild and Severe recovery rates.
- Simpson's Paradox: a treatment can be better in every subgroup but look worse after the groups are combined.
- Randomization: both treatments get the same Mild/Severe mixture, removing the imbalance that causes the reversal.

2. Approach

Use one simple rule throughout:

$$\text{Overall recovery} = (\text{Mild recovery} \times \text{Mild proportion}) + (\text{Severe recovery} \times \text{Severe proportion})$$

First calculate Treatment A's overall recovery in terms of alpha. Then calculate Treatment B's numerical overall recovery. Compare the two to find the alpha values that create the reversal. Finally, repeat the

comparison when both treatments have the same Mild proportion p_M in a randomized trial.

3. Step-by-Step Solution

Two treatments: A and B

Two types of patients:

- M = Mild
- S = Severe
- R = Recovery

Part (a): Overall recovery rate of Treatment A

Let alpha be the proportion of Mild patients receiving Treatment A:

$$P(Z=M \mid A) = \text{alpha}$$

$$P(Z=S \mid A) = 1 - \text{alpha}$$

Treatment A has 90% recovery for Mild patients and 50% recovery for Severe patients.

$P(R|A) = P(R|A,M)P(M|A) + P(R|A,S)P(S|A)$

$$P(R \mid T=A) = 0.90(\text{alpha}) + 0.50(1-\text{alpha})$$

Expand the expression:

$$P(R \mid A) = 0.90\text{alpha} + 0.50 - 0.50\text{alpha}$$

$$P(R \mid A) = 0.50 + 0.40\text{alpha}$$

Answer (a): P(R | A) = 0.50 + 0.40alpha.

Part (b): Overall recovery rate of Treatment B

Treatment B has 80% Mild patients and 20% Severe patients in the numerical setup.

$$P(Z=M \mid B) = 0.80, P(Z=S \mid B) = 0.20$$

Treatment B recovers 80% of Mild patients and 30% of Severe patients.

$$P(R \mid B) = 0.80(0.80) + 0.30(0.20)$$

$$P(R \mid B) = 0.64 + 0.06$$

$$P(R \mid B) = 0.70$$

Answer (b): The aggregate recovery rate of Treatment B is 0.70 = 70%.

Part (c): Exact alpha interval for Simpson's Paradox

First notice that Treatment A is better in each subgroup:

$$\text{Mild: } 0.90 > 0.80$$

$$\text{Severe: } 0.50 > 0.30$$

Simpson's Paradox occurs when Treatment A nevertheless looks worse overall:

$$P(R \mid A) < P(R \mid B)$$

Substitute the overall recovery rates found in parts (a) and (b):

$$0.50 + 0.40\text{alpha} < 0.70$$

Subtract 0.50 from both sides:

$$0.40\text{alpha} < 0.20$$

Divide by 0.40:

$$\text{alpha} < 0.50$$

Because alpha is a probability, it must lie between 0 and 1.

$$\text{Simpson's Paradox occurs for } 0 \leq \text{alpha} < 0.50$$

Answer (c): Simpson's Paradox occurs for 0 ≤ alpha < 0.50 (or 0 < alpha < 0.50 if alpha is restricted to strictly positive proportions).

Part (d): Why a randomized controlled trial prevents the reversal

In an RCT, both treatments have the same Mild-patient proportion p_M:

$$P(Z=M \mid A) = P(Z=M \mid B) = p_M$$

$$P(Z=S \mid A) = P(Z=S \mid B) = 1 - p_M$$

Treatment A overall recovery is:

$$P(R \mid A) = 0.90p_M + 0.50(1-p_M)$$

$$P(R \mid A) = 0.50 + 0.40p_M$$

Treatment B overall recovery is:

$$P(R \mid B) = 0.80p_M + 0.30(1-p_M)$$

$$P(R \mid B) = 0.30 + 0.50p_M$$

Now subtract Treatment B from Treatment A:

$$P(R \mid A) - P(R \mid B)$$

$$= (0.50 + 0.40p_M) - (0.30 + 0.50p_M)$$

$$= 0.20 - 0.10p_M$$

For any valid p_M in the interval (0,1):

$$0.10 < 0.20 - 0.10p_M < 0.20$$

So the difference is always positive. Therefore Treatment A always has a higher aggregate recovery rate than Treatment B when both treatments use the same Mild/Severe proportions.

$$P(R \mid A) > P(R \mid B)$$

Answer (d): Randomization prevents the Simpson's Paradox reversal because both treatments are compared using the same patient mix, and A remains better overall for every p_M in (0,1).